

Motor Imagery EEG Signal Classification Scheme Based on Entropy of Intrinsic Mode Function

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Abstract—In this paper, an efficient scheme is proposed for classification of electroencephalogram (EEG) data in to different motor imagery (MI) tasks based on empirical mode decomposition (EMD). The EEG data recorded from each channel is first decomposed into a set of intrinsic mode functions (IMFs) by using the EMD analysis. In view of extracting discriminative information from the EEG signals corresponding to different MI tasks, we propose to utilize the entropy of band-limited IMF. Instead of considering all IMFs or first few IMFs, in the proposed method only the first IMF is chosen because of its low variance. In order to reduce the dimension of the feature vector consisting of entropy values from all channels, principal component analysis is performed. For the purpose of classification, train and test datasets are formed as per leave one out cross validation scheme and then linear discriminant analysis (LDA) is carried out. Simulation is performed on publicly available MI dataset IVa of BCI Competition-III to classify the MI data in to two classes, namely right hand and right foot MI tasks. It is observed that the proposed classification scheme not only offers significant reduction in feature dimensionality but also provides satisfactory classification accuracy.

Index Terms—Brain computer interface, classification, electroencephalogram, empirical mode decomposition, feature extraction, linear discriminant analysis, motor imagery.

I. INTRODUCTION

Electroencephalography (EEG) has become the most popular and widely studied non-invasive brain-computer interface (BCI) due to its excellent temporal resolution, usability and low setup costs. EEG-based BCIs use electrical brain activity in assisting, augmenting, and repairing human cognitive functions. The greatest challenge in BCI systems is to correctly and efficiently identify different EEG signals of different motor imagery (MI) tasks. Recently, MI EEG signal classification has received much attention by several researchers [1]-[4]. With the ease of availability of high quality low cost EEG leads, one major concern now-a-days is the huge amount of data to be handled in case of multi-lead EEG signal analysis. Different spatial filters can be used for EEG channel reduction, where a major drawback is the manual selection of regularized parameter. For example, sparse spatial filter optimization method, introduced in [1], can offer EEG channel reduction at the expense of manual parameter selection. Similarly, the regularized common spatial pattern (CSP) algorithm, reported in [2], involves two regularization parameters which are not selected optimally. Four different approaches for MI task classification using EEG signals are proposed in [3] based on CSP which itself is very sensitive to

noise and often over fits with small training sets. The regularization and aggregation techniques introduced in [4] solve the parameter determination problem by cross-validation method but only in a small sample setting. In [5], least squares support vector machine (LS-SVM) based clustering technique is proposed for feature extraction but the parameters are not selected optimally. Although a high level of cross validation accuracy is reported in this method, a prior knowledge on the class labels is assumed to be available during cross-correlation based feature extraction. Considering the non-stationary nature of EEG signal, empirical mode decomposition (EMD) is also applied for EEG analysis in different applications [6]-[8]. Recently in [8], the potential of EMD analysis in MI data handling is discussed with a note that relevant features must be carefully selected for getting better accuracy.

The objective of this paper is to develop a robust classification scheme based on EMD analysis of the EEG signal corresponding to different MI tasks. The EMD operation is first performed on EEG data of each channel to obtain a set of intrinsic mode functions (IMFs). Next, the entropy of band pass filtered first IMF of each channel data is computed and a feature vector is formed. Because of a large number of channels, the feature dimension becomes large and thus PCA based dimension reduction is performed. Finally, linear discriminant analysis is carried out for classifying the EEG data into two MI tasks. Performance of the proposed method is validated by using a publicly available MI EEG dataset IVa [9]-[10] of BCI Competition III.

II. DATA ACQUISITION

In this paper, most widely used publicly available EEG dataset IVa of BCI Competition III is considered for the purpose of classification [9]-[10]. This dataset was recorded from five healthy subjects labeled aa, al, av, aw, and ay. In this dataset, EEG signals from 118 electrodes were collected based on the international 10/20 system. There were 280 trials for each subject, namely 140 trials for each task per subject. During the each trial, each subject was required to perform either of the two (right hand and right foot for IVa) MI tasks for 3.5 seconds. The down-sampled data at 100 Hz is generally used although the original sampling rate was 1000 Hz. Duration of the recording of the MI task is 3.5 seconds.

III. PROPOSED METHOD

A. Empirical Mode Decomposition (EMD)

The EMD is a time-frequency analysis tool for nonlinear and nonstationary signals [6]-[7]. It is a fully data driven technique

with which any complicated data set can be decomposed into a finite and often small number of intrinsic mode functions (IMF). A function is considered to be an IMF if it satisfies two conditions; first, in the whole data set, the number of local extrema and that of zero crossings must be equal to each other or different by at most one and second, at any point, the mean value of the envelope defined by the local maxima and that defined by the local minima should be zero. The systematic approach to decompose the data $x[n]$ into IMFs, known as the “sifting” process, can be briefly summarized as follows [7]:

- Identify all local extrema of the signal. Next, interpolate maxima to define an upper envelope E_U and interpolate minima to define a lower envelope E_L . Set $h_1[n] = x[n] - m_1$; $m_1 = \{E_L[n] + E_U[n]\}/2$.
- Iterate on h_1 until a stopping criterion is satisfied. Upon stopping one has an IMF C_1 . The extraction of subsequent IMFs works by iterating on $r_1 = x - C_1$ and so on until an outer loop stopping criterion is satisfied. The result is $x[n] = C_1 + C_2 + \dots + C_N + r_N$.
- Let after w cycles of operation, $h_w[n] = h_{w-1}[n] - m_w$ becomes an IMF. Repeating the above process for L times, L no. of IMFs is obtained along with the final residue $r_L[t]$. A popular stopping criteria for the sifting process is to have the value of standard difference (SD) within a threshold as:

$$SD = \sum_{n=1}^N \frac{|h_{w-1}[n] - h_w[n]|^2}{h_w[n]^2}, \quad (1)$$

where w and $w-1$ are index terms indicating two consecutive sifting processes. Thus the decomposition process is stopped since $r_L[n]$ becomes a monotonic function from which no more IMF can be extracted. To this end, for L level of decomposition, $x[n]$ can be reconstructed as

$$x[n] = \sum_{k=1}^L c_k[n] + r_L[n] \quad (2)$$

Here $c_1, c_2, \dots, c_L$ represent the IMFs. In this work, EEG data of each channel is decomposed into 8 IMFs.

For the purpose of feature extraction, it would be very difficult to consider all IMFs, which will not only increase the computational burden but also reduce the feature consistency. In many cases, first few IMFs are chosen. However, in this paper, it is decided to consider only one of the eight IMFs. For the purpose of analysis, first, the variance of the IMF data pattern is measured. In Fig. 1(a), the variance of the first and second IMFs are plotted for different trials considering a fixed channel (channel no. 1). In a similar way in Fig. 1(b), the variance of the first and second IMFs are plotted for different channels considering a fixed trial (trial no. 1). In Fig. 1(c), variation of energy over different trials for a fixed channel is shown for the first three IMFs. It can be clearly observed that the variance of the first IMF is lower in most of the cases in comparison to that of the second IMF and its energy is also comparatively lower. Hence, in the proposed method only the first IMF is considered for feature extraction.

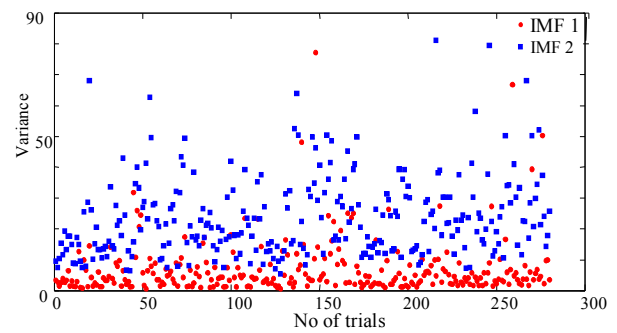


Fig. 1(a). Variance of the first and second IMFs over different trials for a fixed channel.

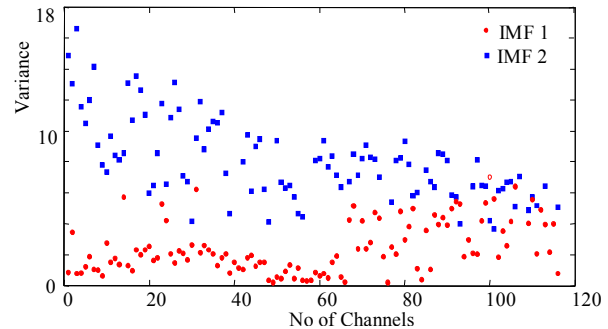


Fig. 1(b). Variance of the first and second IMFs over different channels for a fixed trial.

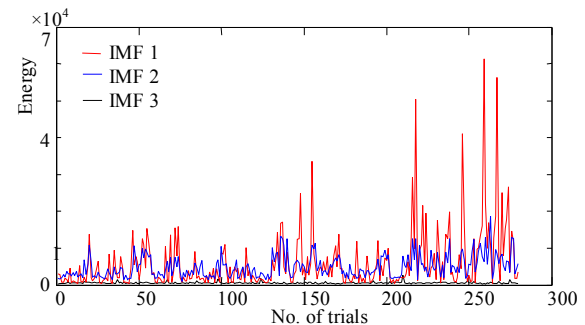


Fig 1(c). Variation of energy over different trials for fixed channel for first three IMFs.

Next the frequency content of the first IMF is investigated. In low frequency region, it is very difficult to discriminate the data for two different MI tasks. Alternatively, it is expected to find more distinguishable characteristics in the high frequency region for the first IMF. It is to be mentioned that such a scenario may not be true for other IMFs. Hence, the first IMF is band limited using a high pass filter. The frequency spectra of a sample first IMF is shown without and with band limitation in Figs. 2(a) and 2(b), respectively. It is observed that the spectra exhibit more distinguishable information in the high frequency range as expected.

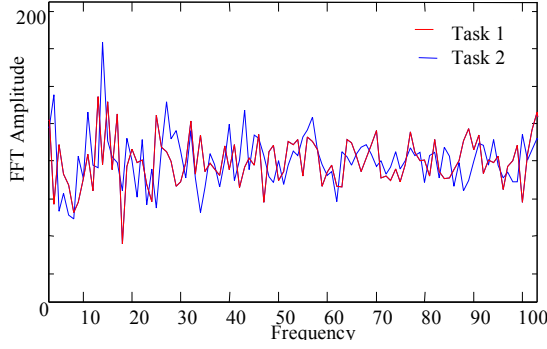


Fig. 2(a). Frequency spectra corresponding to the first IMF of two tasks without using band limiting operation

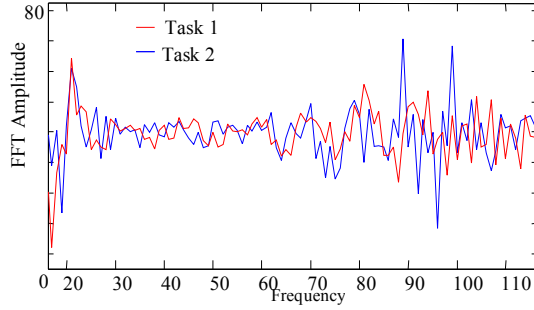


Fig. 2(b). Frequency spectra corresponding to the first IMF of two tasks after band limiting operation.

B. Proposed Feature: Entropy of Band-limited IMF

Entropy of a discrete random variable X with $(N+1)$ number of possible values $\{x_0, x_1, x_2, \dots, x_N\}$ is defined as

$$H(X) = E(I(X)), \quad (3)$$

where $E(\cdot)$ denotes the expectation operator and $I(X)$ represents the information content. For a particular value x_i of X , the information content can be expressed as

$$I(X) = -\log_2(p(x_i)), \quad (4)$$

where $p(x_i)$ denotes the probability of occurrence of x_i . Using (4), the entropy in (3) can be rewritten as

$$H(X) = -\sum_{i=0}^N p(x_i) \times \log_2(p(x_i)). \quad (5)$$

Here $p(x_i) = n_i/N$ and where n_i is the number of occurrence of coefficients in a bin in proximity of x_i value among the N number of values, i.e. $\sum_i n_i = N$. In the proposed method, entropy of the band limited first IMF is considered as feature. In Fig. 3(a), variation of entropy values over different trials for a fixed channel (channel no. 1) is shown. Next in Fig. 3(b), variation of entropy values over different channels for a fixed trial (trial no. 1) is presented. From these figures, it is difficult to comment on the quality of the extracted feature. In order to improve the feature quality, instead of considering the entire duration of an IMF in entropy calculation, we propose to compute entropy in a small duration. The band limited first IMF is windowed into 10 overlapping frames with 50% overlap and the entropy is calculated in each window. An average value of entropy over then frames is considered as a

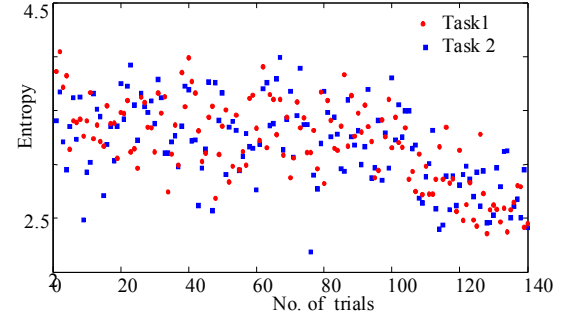


Fig 3(a). Variation of entropy values over different trials for fixed channel (channel 1).

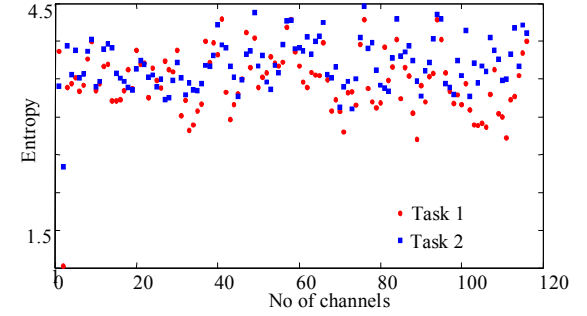


Fig 3(b). Variation of entropy values over different channels for fixed trial (trial 1).

desired feature. It is observed that instead of computing entropy once on the whole duration, computing it in all small windows and then considering the average value significantly enhances the feature quality. Entropy values obtained from all the channels are used to form the desired feature vector.

C. Dimensionality Reduction Using PCA

As in the feature vector, entropy values from all channel are considered, the feature dimension becomes large. Next, principal component analysis (PCA) is carried on to reduce the feature dimension. The PCA is a very well-known and efficient orthogonal linear transformation [11]. It transforms the original p -dimensional feature vector into the L -dimensional linear subspace spanned by the leading eigenvectors of the covariance matrix of feature vector in each cluster ($L < p$). PCA is theoretically the optimum transform for given data in the least square sense. For a data matrix $\mathbf{X}^T$ with zero empirical mean, PCA transformation is given by

$$\mathbf{Y}^T = \mathbf{X}^T \mathbf{W} = \mathbf{V} \mathbf{\Sigma}^T \quad (6)$$

where the matrix $\mathbf{\Sigma}$ is a $m \times n$ diagonal matrix with nonnegative real numbers on the diagonal and $\mathbf{W} \mathbf{\Sigma} \mathbf{V}^T$ is the singular value decomposition of $\mathbf{X}$. Implementation of PCA on the derived feature space not only reduces the feature dimension it also helps in obtaining better feature characteristics. In order to demonstrate the effectiveness of using the PCA, feature values corresponding to two tasks obtained after performing the PCA are shown in Fig. 4. It can be observed that the proposed features after performing PCA exhibit better separability between the two tasks.

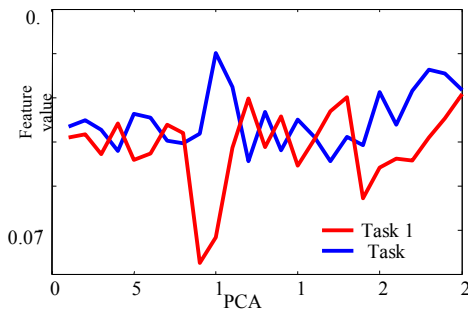


Fig 4. Comparison of feature values obtained after PCA.

D. Classification

In the proposed method, linear discriminant analysis (LDA) is used to classify the testing EEG signals among two different classes (tasks). In LDA, the objective function is based on determining a projection direction $\mathbf{w}$ to maximize the Fisher's discriminant defined as

$$J(\mathbf{w}) = \frac{\mathbf{w}^T \mathbf{S}_b \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}} \quad (7)$$

where $\mathbf{S}_w$ and $\mathbf{S}_b$ are within- and between-class scatter matrices, respectively [12]. Classification performance is measured using leave-one-out cross validation where the test is performed for all 280 datasets (both training and testing) for each subject.

IV. SIMULATION RESULTS

For the purpose of testing with MI EEG data, sample values are first imported into EEGLAB GUI [13]. A reference electrode TP10 is chosen during data import. Without a reference electrode, it is possible that minimum level of 40 dB noise is accumulated in channel information [13], [14]. Typical references in the EEG recording are TP10, Cz or channel on the nose tip. EEGLAB is also used to separate frames of two classes from a jumbled dataset. In the band limiting operation on the first IMF, high frequency band is selected as 70-120 Hz. After PCA the feature dimension is reduced to 20. Finally, features obtained after PCA is used in the LDA to classify the two tasks via leave one out cross validation. Accuracy is measured based on the correctly identified number of trials among all 280 trials. In Table I, comparative results of the classification accuracy are presented for dataset IVa for the five subjects as well as the overall mean accuracy [1], [14]. It is clearly observed that the performance of the proposed method is significantly better in cases of subject av and aw, which is very promising in subject specific applications. However, the performance in case of subject al is relatively poor and needs further investigation. Overall the proposed methodology provides competitive average classification accuracy compared to all the reported algorithms.

V. CONCLUSION

A simple but efficient MI classification scheme is developed based on EMD analysis. It is found that use of only the first IMF can provide satisfactory classification performance when entropy of the band-limited IMF is considered as the feature. Because of considering only the entropy value of the selected IMF, the feature dimension is drastically reduced. Moreover, use of PCA provides further control on feature dimension reduction as well as significant improvement in the derived feature quality. Because of the

TABLE I. PERFORMANCE COMPARISON OF DIFFERENT METHODS

Method	Classification Accuracy (%)					
	aa	al	av	aw	ay	Average
Proposed	72.64	75.17	73.36	88.51	83.59	78.65
CSP [14]	66.96	89.29	52.25	47.77	52.38	61.79
CSSP [14]	79.46	92.86	52.55	91.52	51.59	73.60
BSSFO [14]	79.46	94.64	57.65	91.96	53.57	75.46
FBCSP [14]	69.64	80.36	47.96	55.36	48.41	60.35
DCSP [14]	69.64	82.14	54.08	50.89	48.41	61.03
SSRCSP [1]	70.54	96.43	53.57	71.88	75.39	73.56
TRCSP [1]	71.43	96.43	63.27	71.88	86.9	77.98
WTRCSP [1]	69.64	98.21	54.59	71.88	85.32	75.93
SRCSPP[1]	72.32	96.43	60.2	77.68	86.5	78.63

quality of the extracted features, even simple LDA based classifier can provide satisfactory classification performance. It is observed that the proposed entropy based method offers a very satisfactory classification performance for all subjects in comparison to some of the existing methods.

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